

AI Tools for Study & Work

Detailed Chapter Notes & Worked Examples - Volume 2

This pack develops practical understanding through explanations, worked examples, lab tasks and review questions. Complete the matching Foundation Pack first.

Chapter	Topic
1	Prompt engineering
2	Verification workflow
3	Office productivity
4	Responsible project

Chapter 1: Prompt engineering

Concept notes

A useful prompt states role, goal, context, constraints, source material and output format. Iteration should correct a specific weakness rather than merely asking for better output.

Worked example

Prompt: Explain photosynthesis to a Class 8 learner in 150 words, then give five MCQs with answers; use only the supplied textbook passage.

Lab assignment

Write three prompts for explanation, drafting and data analysis; test and revise each.

Review questions

1. What makes context useful?
2. Why specify output format?
3. What should be verified?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 2: Verification workflow

Concept notes

AI output is a draft, not evidence. Separate factual claims, calculations and recommendations; verify each through primary or authoritative sources and record uncertainty.

Worked example

For a scholarship answer, check eligibility on the official portal, deadline on the current notice and calculation independently.

Lab assignment

Fact-check an AI response and create a claim-source-verdict table.

Review questions

1. What is hallucination?
2. Why prefer primary sources?
3. How should uncertainty be shown?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 3: Office productivity

Concept notes

AI can outline documents, rewrite tone, suggest formulas and create presentation structures. Do not upload confidential student, identity, payment or password data.

Worked example

Use fictional data to draft a fee reminder, create an Excel formula explanation and outline a six-slide orientation.

Lab assignment

Produce the three outputs and document human corrections made.

Review questions

1. Which data is sensitive?
2. Why use fictional data for practice?
3. What requires human approval?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 4: Responsible project

Concept notes

Assess usefulness, accuracy, privacy, fairness and ownership. Keep a prompt log and disclose material AI assistance where appropriate.

Worked example

Build a revision-plan assistant using non-sensitive inputs and a checklist that rejects unsupported claims.

Lab assignment

Demonstrate the workflow with before/after output and risk controls.

Review questions

1. What is bias?
2. When should AI use be disclosed?
3. Who remains accountable?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Final Applied Assignment

Create one original project that combines all four chapters of AI Tools for Study & Work. Use realistic but non-sensitive sample data. Submit planning notes, source files, final output, test evidence and a one-page reflection.

Assessment rubric

Area	Marks
Concept accuracy	25
Working practical output	30
Validation and testing	20
File organisation and presentation	15
Original explanation	10

Student declaration

I confirm that the submitted work is my own practice. I have not included passwords, OTPs, identity documents or confidential student records.

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